Practice workbook · Pick the right data type for a value

Recognising which primitive data type suits a value or variable, giving example values for a type, and stating the values a Boolean can hold.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-01

1. Question 1 · part e · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part e only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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1. Question 1 · part e · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part e only.

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(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

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(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

1. Question 1 · part e · 2 marks · Working and answer

Working and answer

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2. Question 1 · part a · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part a only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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2. Question 1 · part a · 1 mark · Working and answer

Working and answer

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3. Question 2 · part b.iii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b.iii only.

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2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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3. Question 2 · part b.iii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b.iii only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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P 6 1 8 8 1 R A 0 6 1 6

3. Question 2 · part b.iii · 2 marks · Working and answer

Working and answer

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4. Question 1 · part b · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part b only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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4. Question 1 · part b · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part b only.

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(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

4. Question 1 · part b · 1 mark · Working and answer

Working and answer

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5. Question 1 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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5. Question 1 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

(d) Describe **one** difference between the data used in boundary testing and the data used in erroneous testing.

(2)

(e) A program should output the value 2

However, there is an error in the code and the actual output is 4

Name this type of error.

(1)

(f) Open **Q01f** in the code editor.

The program should allow the input of the length of the side of a square and output the area of the square.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(3)

(g) A program is needed that will accept the input of a letter and compare it with a stored letter.

It will check whether the letter comes earlier in the alphabet, later in the alphabet or is the same letter as the stored letter.

It will output the letter and the stored letter with an appropriate message.

Open **Q01g** in the code editor.

Amend the code to complete the if statement used to produce the output.

You must use the structure given in **Q01g** to complete the program.

Do not add any further functionality.

Save your code as **Q01gFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 1 = 14 marks)

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5. Question 1 · part c · 2 marks · Working and answer

Working and answer

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6. Question 2 · part b.i · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part b.i only.

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2 Programs can process numerical data.

(a) Identify the method used to check input data matches set rules.

(1)

- ☒ **A** Abstraction
- ☒ **B** Encryption
- ☒ **C** Iteration
- ☒ **D** Validation

(b) **Figure 2** shows a section from an employee payslip produced by a computer program.

Pay Rate	Hours		Total Pay
£ 18.75	19		£ 356.25
	Tax	20%	£ 71.25
		Income	£ 285.00

Figure 2

(i) Complete the table to give **one** data type shown in **Figure 2** and give an example value.

(2)

Data type	
Example	

(ii) Complete the table to give an input, a process and an output used in the program to produce the payslip.

(3)

Input	
Process	
Output	

6. Question 2 · part b.i · 2 marks · Working and answer

Working and answer

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